

CDE2701A Aspirational Project

**Coffee Extraction Simulation with a Pore
Network Model**

Interim Report



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1. Project overview

Coffee is a beverage enjoyed by many people all around the world and is prepared in many different forms – espresso, moka pot, pourover etc., with vastly different taste profiles depending on personal preference. In fact, even within each of the preparation methods listed above, great variances exist depending on the exact parameters used, leading to differences in the chemistry and hence, the taste. Owing to the pervasiveness of coffee in everyday life, studies have been done since 1957 to optimise the flavour of coffee, first by establishing measurable metrics that could be linked to taste and eventually expanding to finding parameters that could optimise for these metrics.

While good tasting coffee is extremely subjective and not yet fully developed by sensory scientists, there exist at least 2 common parameters that are generally agreed upon to have correlations to taste. The first, total dissolved solids (TDS), is the concentration of dissolved coffee compounds while the second, extraction yield, is the mass fraction of the dry coffee that is extracted. Recommendations originally called for 1.15-1.35% TDS and 18-22% extraction yield (Lockhart 1957), although recent trends have pushed for coffee to be enjoyed towards the upper end of these ranges. As a further note, different preparation methods usually result in fairly different TDS values, such as 1.2-1.7% for pourover and 8-12% for espresso, but the above serves as an average guide across all preparation methods, causing certain methods such as espresso to be neglected (Gagne 2020).

Most studies analysed various brewing parameters' effects on extraction, TDS and other method specific parameters such as temperature at various points in the brewing setup, but a few explored the underlying physics behind the extraction process. Such physics lie within the field of multi-phase flows in porous media, with many different methodologies employed for the express purpose of coffee simulations thus far, such as the Lattice Boltzmann Method (LBM) (Misici, et al. 2005), Smoothed Particle Hydrodynamics (SPH) (Ellero and Navarini 2019) and Direct Pore-scale Numerical Simulations (DPNS) (Giacomini, et al. 2020).

Pore Network Modelling (PNM) has a similarly rich history among the previously mentioned methods and is noted to be a relatively accurate framework for studying multi-

phase flows (Golparvar, et al. 2018), although it does have its drawbacks. PNM consists of a network of pores representing space, responsible for holding the bulk volume of fluids linked by throats, considered as fluid conduits. Conductances, which are a parallel to that in electrical circuits, are assigned to each pore and throat and allow flow to be approximated throughout the network. PNM is known to be relatively accurate in situations involving significant external forces such as gravity but struggles to capture fine details leading to certain phenomena such as fractures within micropores and has not yet been validated for use in multiscale simulations.

PNM has not yet been used for the express purpose of conducting coffee simulations, partly due to advances in availability of computational power which have allowed DPNS to become the preferred method for conducting such studies, as well as being rather simplistic compared to other simulation methods. Despite this, studies acknowledge that there exist distinct stages in the coffee extraction process which require different levels of fidelity (Moroney, Lee, et al. 2015). As such, PNM may still have a purpose as a more efficient model with a lower cost if it can be shown to have sufficient accuracy despite its lower resolution and simplicity.

As such, this project explores the use of pore network models as a framework for the simulation of coffee extraction and attempts to evaluate its accuracy against other simulation methods as listed above. While the results are generalisable to other forms of coffee brewing, the focus lies in pourover coffee brewed by a V60 brewer. This project is open-source and available as a repository at <https://github.com/BearB3ar/coffee-simulations>, or explored in website form at <https://bearb3ar.github.io/coffee-simulations/>.

2. Implementation

Implemented in Python, this simulation utilises 2 main libraries – OpenPNM for its pore network model flow, physics models and advection-diffusion (AD) solver, and Porespy for its pore network generation capability. As an overview, this simulation first generates the geometry, solves for the flow followed by the transport of solute modelled as a single compound via advection and diffusion in discrete time steps, producing outputs of solute concentration and flow pressure throughout the control volume at each time step.

2.1. Initialisation

The simulation first takes in inputs for initialisation of a 3D domain shape in pixels (x, y, z) and target porosity $\rho\%$. The 3D domain shape determines the bounding box dimensions of the physical control volume to be simulated at a voxel size of 10 microns to 1 pixel.

All grinders produce a grind size distribution with a peak around the desired size and a smaller one at a smaller size (Moroney, Connell, et al. 2019), illustrated in Fig 1. The exact size at the smaller peak is generally a function of the grinder and the desired size but is generally less than 100 microns. Coffee particles of size smaller than the smaller peak are known as ‘fines’.

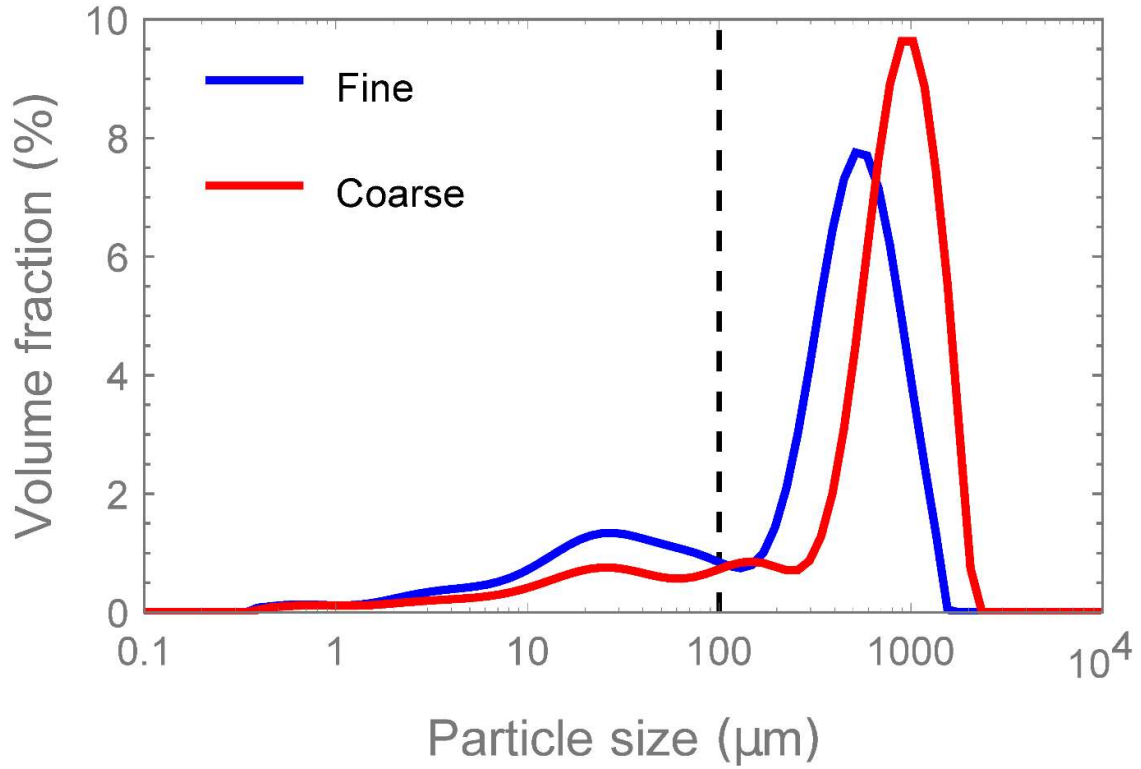


Figure 1: Experimental Data by Moroney et al. 2019 of Grind Size Distribution of a Standard Jacob Kronung Dripfilter Grind for the Fine Grind and Cimbali Burr Grinder Setting #20 for the Coarse Grind

The script captures this general pattern by copying points from the fine grind provided in Figure 1, converting to numerical frequency by dividing by the estimated particle volume and using that frequency distribution to generate spheres of varying sizes in a 3D image up to the desired porosity.

The cuboidal image is trimmed to a cone of side wall angle 60° with respect to the horizontal as constrained by a V60 brewer's geometry, although the geometry can be modified to fit other brewers.

A 'wall effect' increasing the porosity of particle packing near walls is also considered (Seckendorff, et al. 2020). The equation used to describe the maximum porosity boost, Benyahia and O'Neill's relation found in Seckendorff et al.'s research for situations where

$$\lambda = \frac{\text{tube diameter}}{\text{average pore diameter}} \leq 10 \text{ is}$$

$$\epsilon_{boost} = \frac{1.74}{(\lambda + 1.14)^2}$$

This porosity boost is then set to decay exponentially over a decay width of 20% of the maximum control volume diameter. Particles near the walls are probabilistically deleted to achieve this modified porosity.

The distribution of simulated coffee particles for a fine grind after all steps is shown in **Error! Reference source not found.**, and the image of the coffee bed is shown in Figure 3.

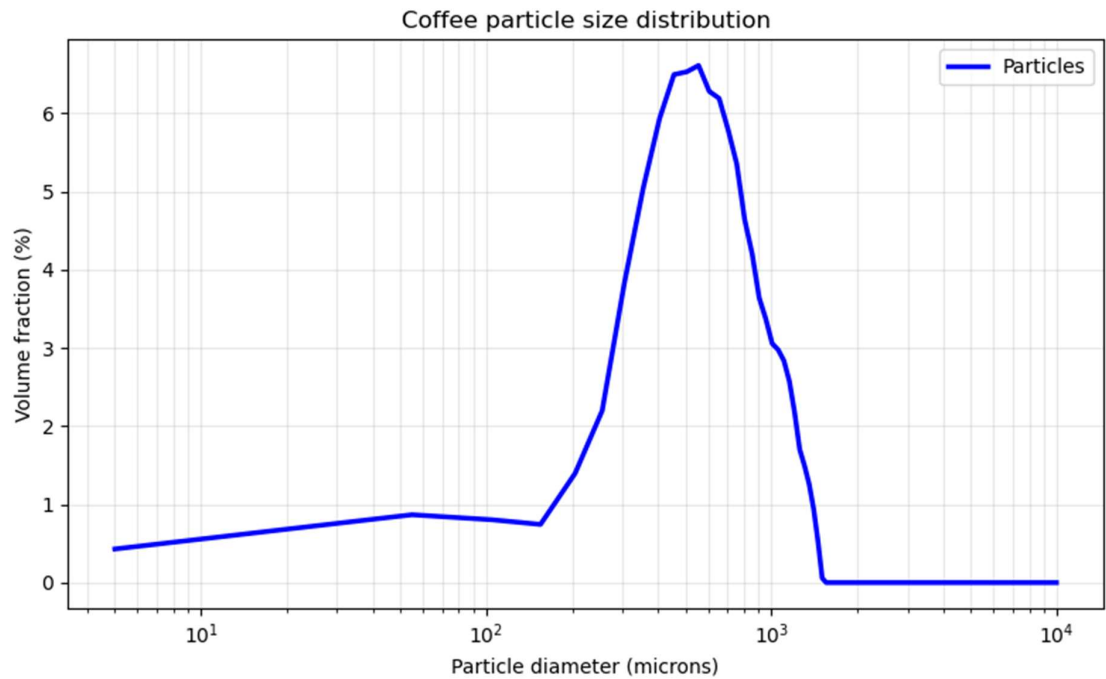


Figure 2: Simulated Grind Distribution for a Fine Grind, Taking Reference from the JK Dripfilter Grind

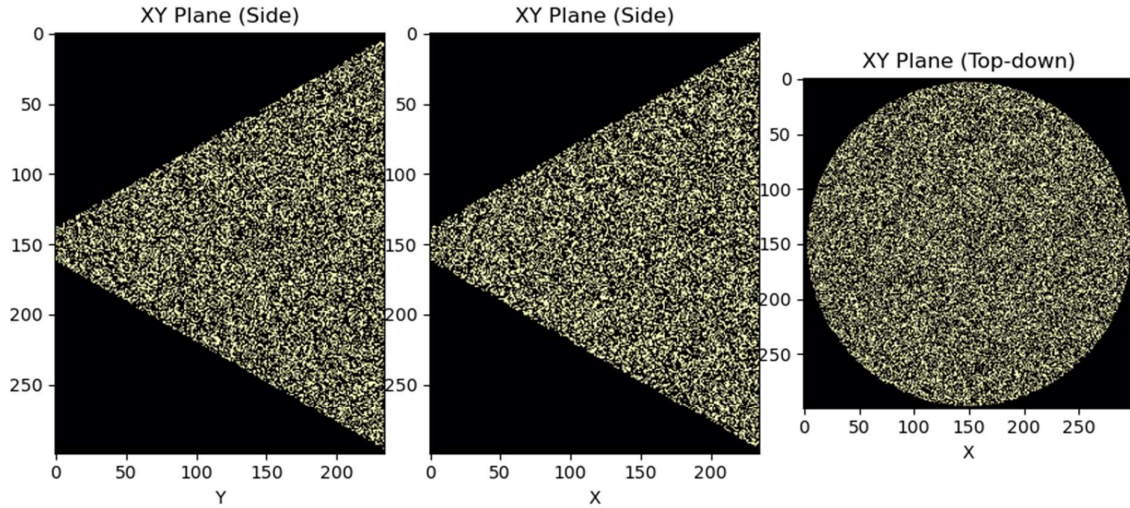


Figure 3: Mid Plane Side Views and Top Down View of the Coffee Bed Image Generated by Porespy

This image is then converted via a watershed algorithm to obtain the final network of pores and throats.

2.2. Physics setup

At this stage, only the positions and directions of pores and throats are known, and hence geometric properties must be applied, such as diameters, cross-sectional areas and volumes, allowing size factors as approximations of geometry to be determined. Together with input temperature which allow water viscosity, diffusivity and thermal conductivity to be calculated, hydraulic, diffusive and thermal conductance factors as approximations of transport properties can also be determined. The specific models used for each approximation are found in the appendix.

2.3. Brew

The simulation then takes in inputs for brew time t in seconds, pour rate q in millilitres per second and time steps n . During each time step, the following steps are performed:

1. Hydraulic flow is solved to produce the pressure and velocity distribution throughout the control volume
2. Thermal flow is solved to produce the temperature distribution throughout the control volume
3. Solutes are added to each pore based on existing concentration in each pore, remaining amount of coffee solute available for extraction and extraction factors
4. Solute transport is solved to produce the concentration distribution throughout the control volume
5. Mobile fines in pores are transported based on probabilistic rule to downstream throats or pores and affect conductance values

2.3.1. Hydraulic flow

The pour rate q is used together with the domain height z to determine the inlet pressure based on hydrostatic and dynamic pressure as shown in the equations below. A circular stream of water of diameter $d = 5\text{mm}$ entering the control volume perpendicularly is assumed, which is a reasonable assumption for water poured from a gooseneck kettle.

$$q = AV = \frac{\pi d^2}{4} V$$

$$P = P_{static} + P_{dynamic} = \rho g z + \frac{1}{2} \rho V^2$$

Combined with a Dirichlet boundary condition of 0 gage pressure at the bottom of the outlet, the pressure and velocity distributions can be found throughout the control volume.

2.3.2. Thermal flow

Using boundary conditions of initial bed temperature and water inlet temperature with thermal conductance and flow velocity distribution as determined above, the thermal flow can be solved to determine the temperature distribution.

2.3.3. Solute introduction

Solutes are modelled as a single compound for simplicity since most compounds extract at linear rates that are nearly identical or factors to each other (Moroney, Connell, et al. 2019). The extraction kinetics have also been found to be a 2 stage process consisting of rapid extraction from the surface of the coffee grains followed by slower extraction at lower concentrations from the interior of the grains (Moroney, Lee, et al. 2015), hence a double extraction model with different extraction factors k_{fast} and k_{slow} are employed to introduce solute into each pore at each time step. The fraction of solute left for extraction and fraction of pore concentration over the saturation concentration also affect how much solute is introduced at each step, as defined by the relation:

$$k_{eff} = k_{fast} \times \frac{m_{fast,remaining}}{m_{fast,initial}} + k_{slow} \times \frac{m_{slow,remaining}}{m_{slow,initial}}$$

The maximum amount of solute extractable was set by tuning the initial concentration to match the equation below.

$$m_i = \text{cone volume} \times \text{actual porosity} \times 765 \times 0.3$$

Where the density of dry coffee is estimated to be 765kg/m³ on average and the amount of solute that can be extracted with typical brew methods is 30% of the bean's dry mass (Gagne 2020).

2.3.4. Solute transport

With the velocity distribution, amount of solute introduced into each pore, diffusive conductance and advective conductance as determined above, the solute transport control volume at each time step can be solved for, producing the concentration distribution.

2.3.5. Fines migration

Fines migration is not technically possible in PNM due to the need to regenerate the control volume with every movement in fines which is computationally expensive and slow, but fines migration forms an integral part of the entire brewing process as it heavily alters residence time throughout the entire brew. As such, this project employs an version of a method used by Misici et al., where particles below a threshold size in a flow above a threshold velocity have a probability of moving to neighbouring larger particles, or may agglomerate into their own large particle (Misici, et al. 2005).

The specific implementation in this project assigns 100 fines to a randomly selected 10% of ‘elite’ pores and 10 fines to all other pores. At the end of every time step, every throat has a probability of collecting fines, defined as:

$$p = u \times 0.4 \times \frac{\text{remaining fines in pores connected to the throat}}{\text{initial fines in pores connected to the throat}}$$

This models the fact that fines move based on the flow velocity and the concentration of fines in the vicinity.

A binomial draw is then conducted with a cap to prevent a throat from drawing more fines than there are available and to collect them evenly from all upstream pores. Throats that are smaller than 100 microns automatically retain all fines that pass through. If more than 100 fines try to pass into a downstream pore, all excess fines are also retained in the throat that tried to pass them. Throat hydraulic conductances are reduced by a factor of $\max\left(1, \frac{\text{throat fines}}{50}\right)$ which represents clogging due to a local accumulation of fines.

When all 100 fines are depleted in each of the ‘elite’ pores, their diameter expands by a factor of 1.5, representing a significant growth in the gap between particles due to significant local fines migration. The ensuing recalculation of physics models also increases the pore hydraulic conductance.

3. Development process

The model was built by first establishing a basic model with only a static AD solver on a small control volume and a normal particle distribution of only the target grind size. Features were then explored and added over time to capture as much complexity as possible. Lastly, all parameters were tuned, using common values found in reality for physical values and a sweep of all coefficients that were declared as approximations, such as the extraction factors k_{fast} and k_{slow} , fraction of solute that is fast f_{fast} and saturation concentration c_{sat} .

The first feature developed from the initial model was a switch from a generic image to network algorithm to a more precise watershed algorithm.

The next introduced multiple solutes each with their own extraction parameters which would have allowed for very fine detail in how specific compounds are extracted at different stages for the brew. However, using a Fickian Diffusion algorithm or similar to model extraction exactly for multiple compounds would've been computationally expensive, so an extraction factor k was employed instead. But ultimately, since it would have been overly difficult to find real data to fit k with for multiple compounds in such a specific situation as well as an inherent loss in detail with an approximation factor for extraction rate, this feature was dropped for a simpler, unified solute model.

A transient AD solver was explored next for its capability to do time stepping dynamically but was ultimately found to be unsuitable due to numerical instability with the dynamic time stepping algorithm.

Next, the pore size distribution was changed from a bimodal to 2 lognormal distributions combined to capture the distribution of gaps between coffee particles more accurately.

The extraction model was also changed from a constant term k to a linear model that depended on the existing concentration in the pore as well as putting a realistic limit on the amount of solute that can be extracted in total.

A swelling model was also explored at this stage owing to frequent anecdotes from coffee professionals that swelling occurred during the brew. However, Maille et al. found that

coffee particles do not swell appreciably in a regular time frame of coffee brewing across a range of coffees (Maille, et al. 2021). As such, this feature was also deprecated.

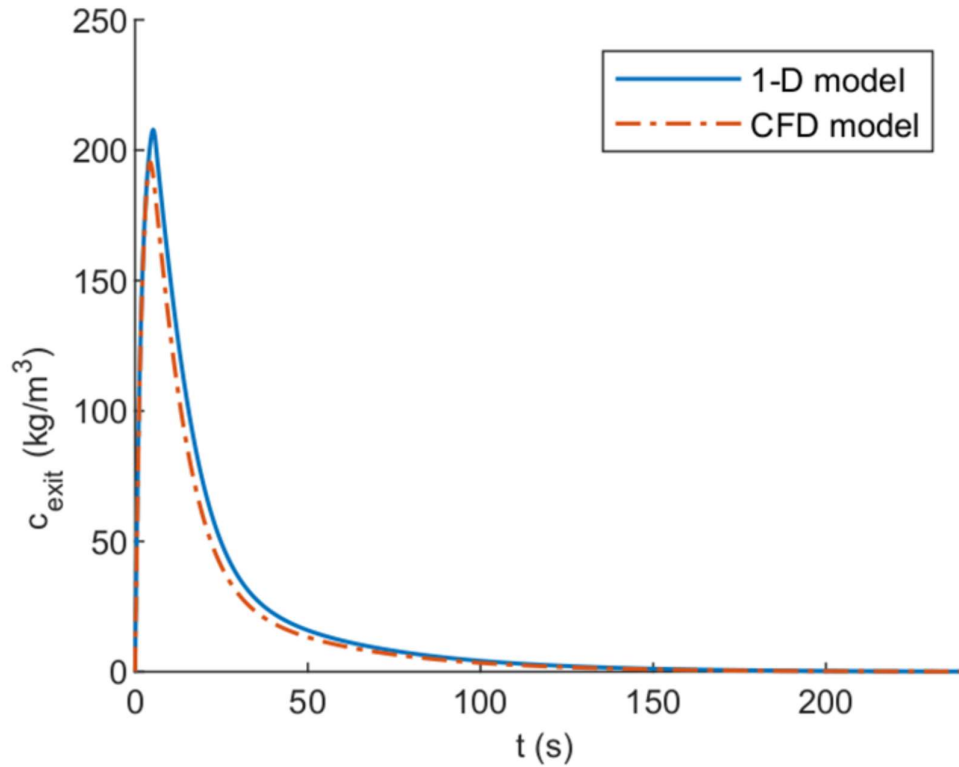
The coffee bed was then trimmed to a cone shape to match the V60's geometry specifically and the 'wall effect', which affects the packing of substances near walls, was considered by increasing the porosity by a decaying amount from distance from the walls.

Next, temperature dependence was added due to the significance of changes in water viscosity in the Stokes flow regime over a wide range of temperatures, especially in the first few seconds where temperatures rapidly rise. An AD solver was also added to dynamically solve for the temperature distribution at each time step.

Finally, the single extraction factor model was changed to the double extraction model to increase realism without a significant increase in computational complexity.

Originally, simulations were conducted to sweep through possible combinations of the extraction factors k_{fast} and k_{slow} , fraction of solute that is fast f_{fast} and saturation concentration c_{sat} . The concentration in the simulated beverage was determined and compared against the concentration obtained by an experiment for a cylindrical brewing chamber as originally studied in Moroney et al., 2015. However, testing the tuned parameters on the experiments from Moroney et al., 2019 which were done on a conical brewing chamber with side angle 60 degrees as appropriate for pourovers produced unrealistic extraction yield and concentration results.

As such, the tuning was redone directly on the conical brewing chamber containing 60g of coffee grounds exposed to an inlet flow rate of 250ml/min as specified by Moroney et al., 2019. Moroney et al.'s experimental results are as shown in **Error! Reference source not found..**



(a) Fine grind

Figure 4: Experimental Data by Moroney et al., 2015 of Coffee Concentration in the Coffee Pot of a Cylindrical Brewing Chamber with a Flow Rate of 250ml/min

The combination of parameters that produced the lowest amount of error across a few categories defined in the cost function below was selected to be the tuned model parameters, which can be found in the appendix. The data was discretised into points in 5 second intervals for the first 30 seconds, then in 10 second intervals until 160 seconds where the exit concentration is approximately 0 for easier comparison.

$$C = 0.5 \times \text{Root Mean Square Error} + 0.3 \times \text{Mean Absolute Error (MAE) of first point} \\ + 0.2 \times \text{MAE of last point} + \text{excess yield out of range } [0.27, 0.33] \times 120$$

The first 3 terms penalise general error directly while the last term penalises unrealistic final extraction yields.

4. Project evaluation

The model is currently being tuned against the experimental results in Moroney et al., 2019. Once complete, the tuned model will be compared against the coarser grind size analysed in Moroney et al.'s 2019 paper which for the same conical brewing chamber.

5. Conclusion

My research attempted to evaluate the use of pore network models (PNM) as a cost effective method of studying multi phase flow in porous media, specifically with the application of coffee extraction, in comparison with more accurate but computationally expensive direct numerical simulations (DNS).

6. References

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7. Appendix

A1. Models used for physical elements

<u>Physical element</u>	<u>Physics model</u>
Pore volume	Sphere
Pore cross sectional area	Sphere
Throat volume	Cuboid
Throat cross sectional area	Pyramids and cuboids
Throat length	Pyramids and cuboids
Throat perimeter	Cuboid
Throat hydraulic size factors	Pyramids and cuboids
Throat diffusive size factors	Pyramids and cuboids
Throat conduit lengths	Pyramids and cuboids

Pyramids and cuboids refers to a setup where pores are assumed to be pyramids and throats are assumed to be cuboids. This mixed setup is used as there is no pyramids model for pores.

A2. Models used for conductance

<u>Conductance</u>	<u>Physics model</u>
Throat hydraulic conductance	Valvatne-Blunt
Throat diffusive conductance	Taylor-Aris
Throat ad-dif conductance	Ad-dif
Throat thermal conductance	Generic thermal

The Valvatne-Blunt model utilises shape factors determined from the geometry physics models to apply Hagen-Poiseuille flow for conduits of various shape classes, allowing it to be generalised across a wider variety of irregular shapes which gaps between coffee particles can take on.

The Taylor-Aris model is more suitable than a generic diffusive model in the high Peclet number regime, of which coffee extraction is a part of.

However, a generic thermal conductance model is employed over a series resistors model for simplicity, as the series resistors model is more suited for situations where pores and throats are of dissimilar sizes.

A3. Algorithms

<u>Solvers</u>	<u>Physics model</u>
Flow	Stokes Flow
Thermal and solute transport	Advection-Diffusion

The use of the Stokes Flow model is appropriate as viscous forces dominate over inertial forces for the flow encountered in a compacted coffee bed. This is supported by the literature employing Darcy’s Law for the specific case of coffee extraction, which is only valid for this type of slow, creeping flow regime (Gagne 2020).

Advection Diffusion is also an appropriate model compared to Fickian Diffusion by capturing bulk fluid motion effects together with diffusive effects as opposed to just concentration gradients.